

Indian Statistical Institute
M.Tech. (CS), Final of Second Semester Examination, 2022-23
Database Management Systems

Full Marks: 50

Date: 11-05-2023

Time: 2.5 Hours

Answer any *five* of the following questions

$5 \times 10 = 50$

1. (a) Cite a single example of a pair of relations, say R_1 and R_2 , that satisfies all three of the following properties together. Justify your answer.

(i) $R_2 \times (R_1 \div R_2) = R_1$

(ii) $R_1 \bowtie R_2 = R_1$

(iii) $R_1 \bowtie R_2 \neq R_1 \times R_2$

- (b) What sort of advantage relational algebra provides *not* being Turing complete?

$(3+3+3)+1$

2. (a) Let there be a relational schema $R = \langle N, O, R, M, A, L \rangle$ that comprises only atomic attributes and holds the set of functional dependencies given as follows.

$$\{N \rightarrow O, N \rightarrow R, RM \rightarrow A, RM \rightarrow L, O \rightarrow A\}.$$

Prove the following things.

- (i) $OR \rightarrow L$ does not hold for R .

- (ii) The largest superkey for R is NORMAL.

- (iii) NO is not a candidate key for R .

- (iv) NM is a candidate key for R .

- (v) The highest normal form satisfied by R is 1NF.

$2+2+2+2+2$

3. (a) Consider the relations $R_1 = \langle A \rangle$, $R_2 = \langle A, B, C \rangle$ and $R_3 = \langle A, C \rangle$ having 100, 200 and 300 tuples, respectively. If the number of distinct entries of A in the tables R_1 , R_2 and R_3 are 20, 10 and 15, respectively, estimate the size of the following query.

$$(\sigma_{A \neq 100}(R_1)) \cup (\pi_A(R_2) - \pi_A(R_3))$$

- (b) Highlight an example where both sort merge join and hash join are equally preferable as the physical plans of join operation.

- (c) How does ordering of joins based on the cardinality of relations influence query tuning?

$6+3+1$

4. Suppose the movies ranked in the Internet Movie Database (popularly known as IMDB) are stored based on the following schema.

MOVIE = $\langle \underline{mid}$: number, title: string, director: string, producer: string, year: date $\rangle$

RATING = $\langle \underline{mid}$: number, rating: number, feedback: string $\rangle$

RATER = $\langle \underline{rid}$: number, mid: number, rname: string, rating: number, rdate: date $\rangle$

The primary keys are underlined in the corresponding tables. Write the following queries in SQL.

- (i) Find all the titles of movies produced by Satyajit Ray after 1955 and before 1960.
- (ii) Find all the titles of movies that have no ratings.
- (iii) Find all the years in chronological order (arranged in the order of time) that have a movie receiving a rating of no less than 8.
- (iv) Find all the raters' names who rated the same movie twice and gave it a higher rating the second time.

2+2+2+4

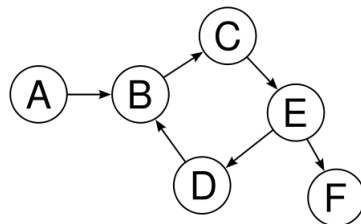
5. (a) Let there be a relation with data items A and B, and 3 different transactions working on them simultaneously at a particular period of time. Suppose the standard two-phase locking protocol is used to manage their concurrency. The entire schedule is shown below.
 - (i) How does the schedule suffer from dirty read operation? Suggest a solution to this problem.
 - (ii) Justify whether it will cause cascading roll back or not. If so, which transactions will be rolled back?

T_1	T_2	T_3	Timestamp
	lock-X(B)		1
lock-X(A)			2
read(A)			3
$A = A + A*0.2$			4
write(A)			5
unlock(A)			6
		lock-X(A)	7
		read(A)	8
		$A = A - 100$	9
		write(A)	10
		unlock(A)	11
	lock-X(A)		12
	read(A)		13
	$A = A + 10$		14
	write(A)		15
	unlock(A)		16

- (b) How does a lock table distinguish the various types and statuses of lock requests saved within it during concurrency control?

((2+2)+3)+3

6. Let there be a precedence graph constructed with the aim of testing conflict serializability of a schedule as shown below. Further consider that no commit instruction is included in the schedule.



Based on the above graph, answer the following questions with appropriate justifications.

- (i) How many transactions are included in the schedule?
- (ii) Is the schedule conflict serializable?
- (iii) Is the schedule view serializable?
- (iv) Can you derive the serializability order of the schedule?
- (v) What is the minimum number of write instructions the schedule includes?

2+2+2+2+2

7. (a) How the mapping cardinality constraints may help to identify the primary keys in the relations constructed from the relationship sets?
- (b) Consider a table that has an attribute having unsorted values and also duplicate entries. Design an indexing strategy to perform range searching efficiently.

5+5
